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M 441 HYDROGEN SYSTEM

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Hydrogen supply pressure to generator cooling system : 3 - 4 kg / cm²
Hydrogen supply pressure at hydrogen plant : 7 kg/cm²

High Pressure storage bottles volume at max 33°C max ambient temp : 750 NM3
Storage tank pressure : 150 kg / cm²
Number of HP storage bottles : determined by contractor

Hydrogen storage sizing shall be determined based 1 (one) fill of one generator but not less than 750 NM3.

The design of the system shall be such as to allow for future expansion when additional turbine generators are installed.

The hydrogen generating equipment shall be located indoor in a building specified.

The building areas shall be considered for classification in accordance with NFPA 70 and NFPA documents referenced there in. Equipment supplied for areas which are classified as hazardous or potentially hazardous shall comply with the requirements of NFPA 70 and referenced NFPA documents. Equipment with standard electrical protection may be installed in separate unclassified areas, as defined by NFPA 70 and NFPA documents reference therein.

3. DETAILED REQUIREMENTS

ELECTROLYZER

Generator module shall include adequate protection against accidental grounding, arcing or electrical shock.

The level of electrolyte or demineralizer shall be automatically controlled.

The equipment shall include the filters and dryers necessary to meet gas purity standards.

Adequate protection against mixing of produced gases and safe atmospheric discharge of oxygen and hydrogen shall be provided.

GAS COMPRESSORS

The compressors shall preferably be diaphragm type with 316 SS diaphragm.

The compressors shall not contaminate the hydrogen with working oil or any other contaminants.

TANKS

The make-up water tank shall comply with the appropriate standards. Water level shall be controlled via solenoid valve with float type level switch assembly mounted at tank top.

VENTS

Vents lines which carry or contain hazardous gas under normal operating conditions shall contain flame arresters or shall be vented at high level in line with safety code.

4. PERFORMANCE TESTING AND GUARANTEE

After equipment is operational The Contractor will run a performance test to ascertain that the equipment is performing properly in every respect. Any deficiencies brought out by this test shall be promptly corrected by Contractor at no cost to Owner.

5. CONTROL AND INTERLOCKS

Control of the compressors shall be selectable from the control panel in a Run, Stop or Auto mode. In the auto mode the selected compressor shall start if the compressor discharge header pressure becomes low or the running compressor is tripped. Low cooling water flow a stage leak, low suction pressure or high discharge pressure shall stop the gas compressors and initiate an alarm. Hydrogen generator shall include critical process monitors which trip the unit under fault conditions ; including high or low gas pressure high electrolyte temperature, low feed water quality, low enclosure air purge flow, high or low electrolyte reservoir level, low electrolyte flow or low electrolyte flow and rectifier power fault. Cause of the fault shall be indicated at generator panel. A generator fault shall initiate a common alarm at annunciator panel. The controls and interlocks shall be modified as necessary to add the equipment needed for future Block additions to provide safe and reliable supply of hydrogen to the plant.